

Alex Rose

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EDUCATION

Massachusetts Institute of Technology
PhD in Control Systems (GPA: 5.0/5.0)

Cambridge, MA
February 2023 – present

Massachusetts Institute of Technology
Master's of Science in Aerospace Engineering (GPA: 5.0/5.0)

Cambridge, MA
June 2021 – February 2023

Thesis: Applying Rotation-Equivariant Deep Learning to Cloud and Road Segmentation in Satellite and Aerial Imagery

Massachusetts Institute of Technology
Bachelor's of Science in Aerospace Engineering (GPA: 5.0/5.0)

Cambridge, MA
August 2017 – June 2021

FELLOWSHIPS & AWARDS

- Draper Scholar (2023-present)
- 2nd place, Frank J. Redd Student Competition at the Small Satellite Conference (2023)
- National Science Foundation Graduate Research Fellowship (2022-present)
- NASA Space Technology Graduate Research Opportunities fellowship (declined) (2022)
- Best presentation award at IEEE RSDM-GeoSci workshop (2022)
- 2022 AIAA Small Satellite Best Paper Award (2022)
- Henry Webb Salisbury Award for superior academic performance (2021)
- Wings Club Foundation Scholarship (2020)
- Intercollegiate Rowing Association All-Academic Team (2019)
- National Merit Scholar (2017)

PREPRINTS

Rose, Alex., Jewison, Christopher., and How, Jonathan P., “PRISM: Efficient and Locally Optimal Probabilistic Planning with Reachability Guarantees,” *arXiv preprint arXiv:2606.26413* (2026).

Rose, Alex., Aggarwal, Naman., Jewison, Christopher., and How, Jonathan P., “Efficient Probabilistic Planning with Maximum-Coverage Distributionally Robust Backward Reachable Trees,” *arXiv preprint arXiv:2510.04807* (2025).

PEER-REVIEWED PUBLICATIONS

Rober, Nicholas., **Rose, Alex.**, and How, Jonathan P. “GUARDIAN: Safety Filtering for Systems with Perception Models Under Adversarial Attack,” *IEEE Control Systems Letters* (2026).

Rose, Alex., Jewison, Christopher., and How, Jonathan P., “Robust Sampling-Based Covariance Steering for Aerocapture Guidance,” AAS/AIAA Astrodynamics Specialist Conference, Boston, Massachusetts, 2025.

Rose, Alex., Aggarwal, Naman., Jewison, Christopher., and How, Jonathan P., “REVISE: Robust Probabilistic Motion Planning in a Gaussian Random Field,” in 2025 American Control Conference (ACC), pp. 3323-3328.

IEEE, 2025.

Thomas, Annika., Sonawalla, Aneesa., **Rose, Alex.**, and How, Jonathan P. “GRAND-SLAM: Local Optimization for Globally Consistent Large-Scale Multi-Agent Gaussian SLAM,” *IEEE Robotics and Automation Letters* (2025).

Meredith, Alex. “Evaluating Rotation-Equivariant Deep Learning Models for On-Orbit Cloud Segmentation”, in 37th Annual Small Satellites Conference, Logan, 2023 - **received 2nd place in Frank J. Redd student paper competition.**

Meredith, Alex., Leroy, Stephen., Halperin, Lucy., & Cahoy, Kerri. “Efficient Collocation of GNSS Radio Occultation Soundings with Passive Nadir Microwave Soundings.” *Atmospheric Measurement Techniques* 16, no. 13 (2023): 3345-3361.

McKeen, Patrick., **Meredith, Alex.**, and Cahoy, Kerri., “Trajectory-Planning Attitude Control System for Satellites with Magnetic Attitude Control and One Reaction Wheel,” AAS/AIAA Astrodynamics Specialist Conference, Charlotte, North Carolina, August 7-10, 2022.

Kacker, Shreeyam., **Meredith, Alex.**, Cahoy, Kerri., & Labrèche, Georges. “Machine Learning Image Processing Algorithms Onboard OPS-SAT”, in 36th Annual Small Satellites Conference, Logan, 2022.

Kacker, Shreeyam., **Meredith, Alex.**, Kusters, Joe., Tomio, Hannah., Felt, Violet., & Cahoy, Kerri. “On-orbit rule-based and deep learning image segmentation strategies”, in AIAA SCITECH 2022 Forum, p. 0646. 2022 – **received 2022 AIAA Small Satellite best paper award.**

do Vale Pereira, Paula., Garcia, Madeline., Schroeder, Madeleine., Caldelas, Humberto., Lindsay, Charles., Choi, Alex., Pfrang, Kaila., Gagnon, Amelia., **Meredith, Alex.**, & Cahoy, Kerri. “BeaverCube: Coastal Imaging with VIS/LWIR CubeSats”, in 34th Annual Small Satellites Conference, Logan, 2020.

CONFERENCE PRESENTATIONS

Rose, Alex., Jewison, Christopher., and How, Jonathan P., “Robust Sampling-Based Covariance Steering for Aerocapture Guidance,” at AAS/AIAA Astrodynamics Specialist Conference, 2025.

Rose, Alex., Aggarwal, Naman., Jewison, Christopher., and How, Jonathan P., ”REVISE: Robust Probabilistic Motion Planning in a Gaussian Random Field,” at 2025 American Control Conference (ACC).

Meredith, Alex. “Evaluating Rotation-Equivariant Deep Learning Models for On-Orbit Cloud Segmentation”, at 37th Annual Small Satellites Conference, Logan, 2023 - **received 2nd place in Frank J. Redd student paper competition.**

Meredith, Alex., Leroy, Stephen., and Cahoy, Kerri. “Rotation Method for Efficient Collocation-Finding”, at IEEE GRSS workshop on Remote Sensing Data Management Technologies in GeoScience 2022 – **received 2022 IEEE RSDM-GeoSci best presentation award.**

Kacker, Shreeyam., **Meredith, Alex.**, Kusters, Joe., Tomio, Hannah., Felt, Violet., & Cahoy, Kerri. “On-orbit rule-based and deep learning image segmentation strategies”, at AIAA SCITECH Forum 2022 (co-presented with Shreeyam Kacker).

OUTREACH PRESENTATIONS

Garcia, Madeline., Green, Dansil., and **Meredith, Alex.** “How to Build a Space Mission”, to a middle-school audience visiting MIT for Spark 2023.

Meredith, Alex. “Pursuing graduate research in engineering as a woman”, to BIO3898 (Women in Science), Department of Biology, Seattle Pacific University, 2022 (updated from 2021 presentation).

Meredith, Alex. “Pursuing graduate research in engineering as a woman”, to BIO3898 (Women in Science), Department of Biology, Seattle Pacific University, 2021.

RESEARCH EXPERIENCE

MIT Aerospace Controls Lab

Graduate Research Assistant

Cambridge, MA

October 2023 – present

- Thoroughly tested novel belief space planning algorithms in simulation, and demonstrated them on hardware (quadrotor flying in cluttered environment)
- Created local optimization algorithm for reducing cost of feasible paths in belief space
- Derived new reachability conditions in belief space, and leveraged these conditions to develop an uncertainty-aware motion planning algorithm with finite-time reachability guarantees
- Developed maximally sample-efficient belief space planning algorithm and proved this algorithm maximizes backward reachability
- Developed robust sampling-based covariance steering algorithm for aerocapture guidance

MIT STARLab

Graduate Research Assistant

Cambridge, MA

June 2021 – October 2023

- Developed fast algorithms that exploit orbital symmetry for colocating satellite data from microwave soundings and radio occultations
- Designed convolutional neural networks (CNNs) to identify clouds in satellite images from visible-spectrum and long-wave infrared cameras
- Optimized planning and trajectory optimization algorithms for underactuated CubeSats using magnetorquer-only control

MIT STARLab

Undergraduate Researcher

Cambridge, MA

June 2020 – June 2021

- Designed and wrote Extended Kalman Filter for CubeSat attitude determination in MATLAB
- Developed physics-based sandbox environment for testing attitude determination and control algorithms using Simulink
- Worked on extending existing magnetorquer-only control algorithms to handle predictable disturbance torques from the space environment and uneven thrust from CubeSat propulsion systems

WORK EXPERIENCE

E-Space

Control Systems Engineering Intern

Beverly, MA

June 2021 – August 2021

- Developed Python simulation modeling orbital environment, disturbance torques, actuators, and attitude determination sensors to design & evaluate magnetorquer-only (underactuated) attitude control system
- Wrote, tested and speed-optimized nonlinear optimization algorithm for planning underactuated slews in C++
- Presented my work on E-Space’s attitude control system to VCs who later invested \$50M in the company

Analytical Graphics, Inc.

Engineering Intern

Exton, PA

June 2020 – August 2020

- Designed CubeSat power system, attitude control system, and optical communications payload and modeled these subsystems using Systems Tool Kit (STK) and STK partner products

- Worked with two other interns to design a CubeSat communications constellation for disaster-relief workers and modeled it in STK
- Completed STK Grandmaster Certification & STK Master Integration Certification

Microsoft

Redmond, WA

Software Engineering Intern

June 2019 – August 2019

- Web-scraped Twitter and did machine learning on datasets of scraped tweets to identify cybersecurity threats in real-time using Python and TensorFlow
- Designed and implemented an internal website using .NET framework to display tweets representing cyberthreat and other cybersecurity-related data in an easy-to-use interface

Explorer Intern

June 2018 – August 2018

- Worked with two other interns to design and implement a data monitoring dashboard in PowerBI, using full Azure stack

SKILLS

Programming Languages: Python, Rust, MATLAB, C++, Java, C, Haskell, JavaScript, HTML/CSS, SQL, Assembly (RISC-V)

Software & Frameworks: STK, ODTK, Keras, Tensorflow, Microsoft Office, Azure, Visual Studio, and PowerBI; BlueSpec